

GPS Data Quality determination and CORS Station Selection to Support Long-Term Monitoring of Ionospheric Anomalies for GBAS

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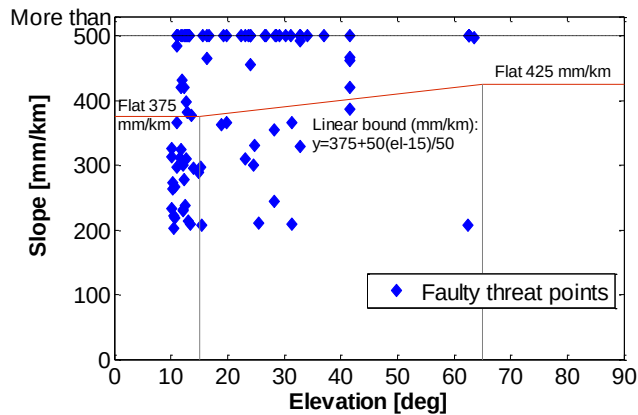


Figure 1. Ionospheric threat space with faulty anomaly candidates before removing stations with poor GPS data quality on 26 May 2012.

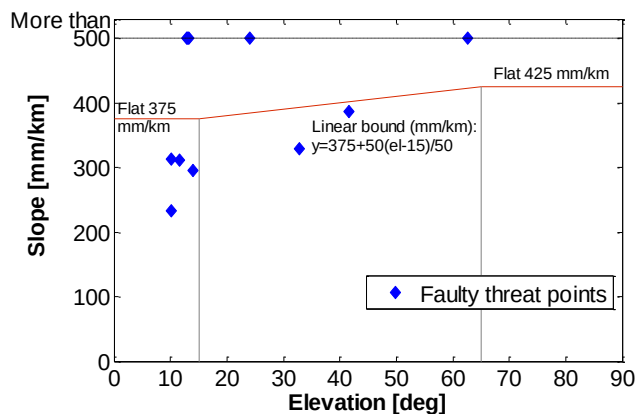


Figure 2. Ionospheric threat space with faulty anomaly candidates after removing stations with poor GPS data quality on 26 May 2012.

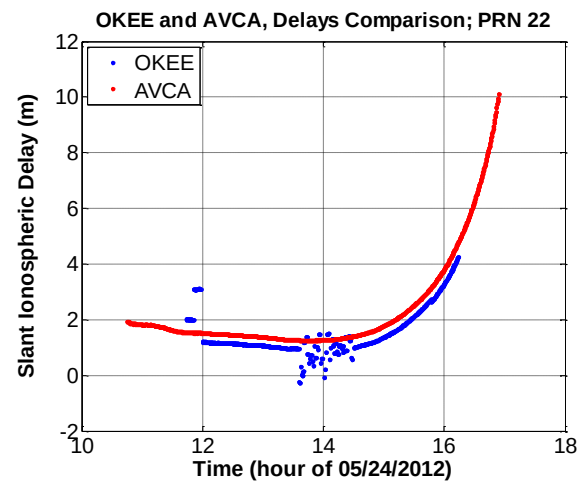


Figure 3. Comparison of dual-frequency slant ionosphere delay estimates for OKEE (red) and AVCA (blue).

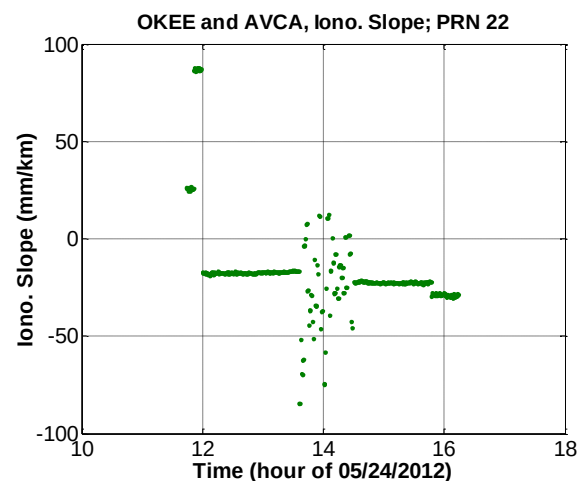


Figure 4. Extremely large faulty ionospheric gradient estimates between OKEE and AVCA tracking PRN 22.

5.1 GPS QUALITY DETERMINATION METHOD

Third, Adaptive filter algorithm are designed to estimate receiver noise on code measurement. After removing the biases of MP1 and MP2 from equation (4), equation (6) is obtained as shown below.

$$MP_i = MP_i + \varepsilon_i \quad (6)$$

As MP_i is the code multipath of the Li frequency, MP_i is highly correlated to the MP_i of the consecutive previous day 오류! 참조 원본을 찾을 수 없습니다.. However, ε_i is the receiver noise on Li code, so it is not correlated to the ε_i of the consecutive previous day. Therefore, according to previous studies, the MP_i of 2 consecutive days can be separated into the correlated part(MP_i) and the uncorrelated part(ε_i) using an adaptive filter 오류! 참조 원본을 찾을 수 없습니다.. For the input of an adaptive filter, a primary input and reference input are necessary, and in this study, the MP_i of the day to be estimated is set as the primary input and the MP_i of the

consecutive previous day is set as the reference input. Then, the output of a filter is calculated using the reference input and weights. The least-mean-square (LMS) algorithm has been used as a computational method to adjust the weights of an FIR filter 오류! 참조 원본을 찾을 수 없습니다..

The output of the adaptive filter is the part in correlation with the reference input. In this study, the MP_i of the primary input (multipath estimates on code measurement) is calculated as the output of the adaptive filter and the estimation error represents ε_i , the receiver noise on code, since the estimation error is the value with MP_i removed from the primary input. Likewise, in order to estimate the receiver noise, ε_i , the correlation between the MP_i of the 2 consecutive days has to exist. However, there are cases where such correlation is not clearly indicated depending on the receiver type and antenna type, and in these cases, the receiver noise in the quality output is presented as N/A.

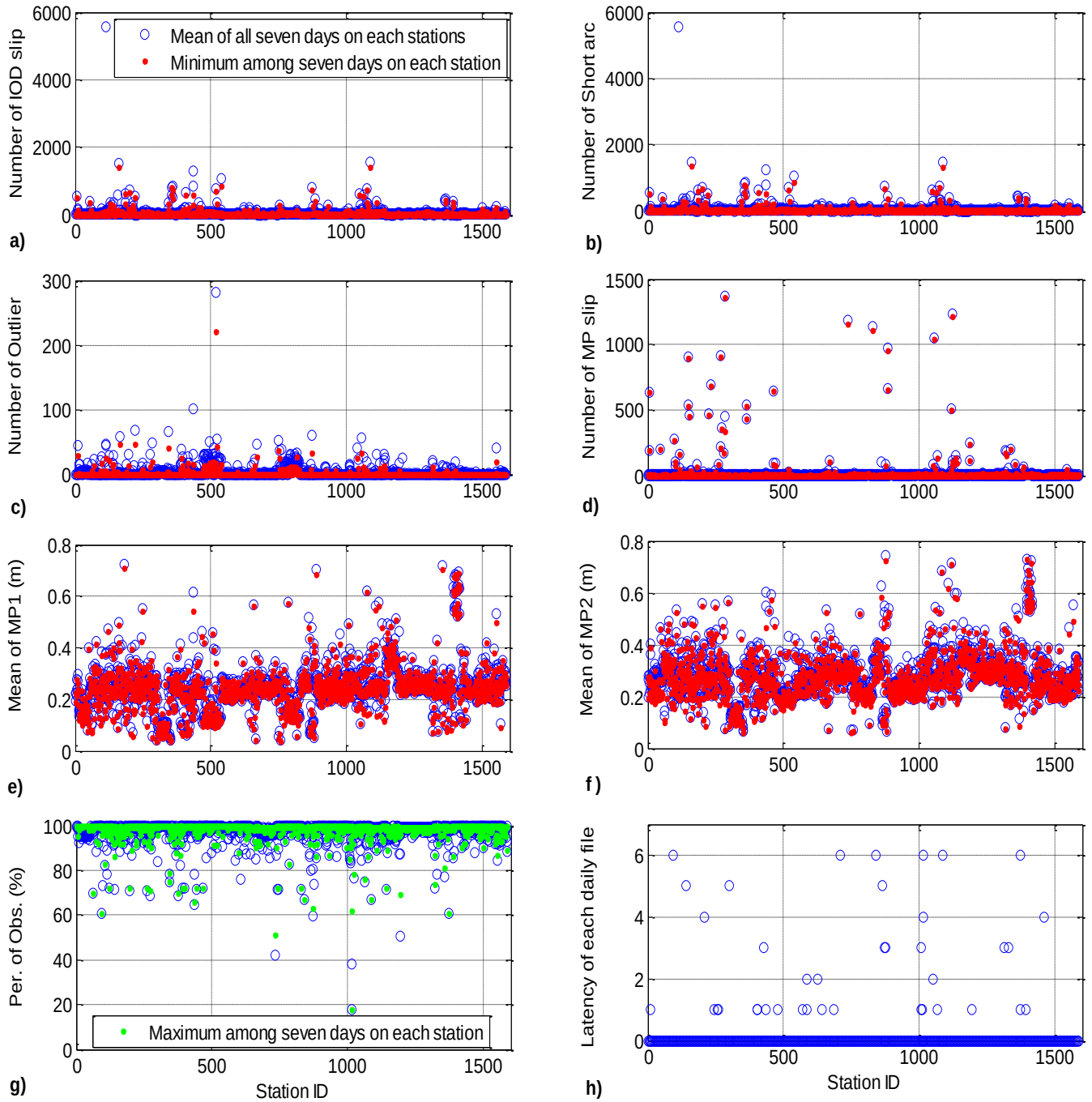


Figure 5. Quality parameters occurred on each station per day: a) number of IOD cycle slips which mean IOD cycle slips of all 7 days and all stations is 37.98, b) number of short arcs which mean short arcs of all 7 days and all stations is 32.74, c) number of outliers which mean outliers of all 7 days and all stations is 3.14, d) number of MP slips which mean MP slips of all 7 days and all stations is 13.24, e) mean of MP1 which average of mean MP1 of all 7 days and all stations is 0.2457 m, f) mean of MP2 which average of mean MP2 of all 7 days and all stations is 0.2826 m, g) percentage of observations which mean percentage of observations of all 7 days and all stations is 97.19%, and h) latency of each daily file for 7days.

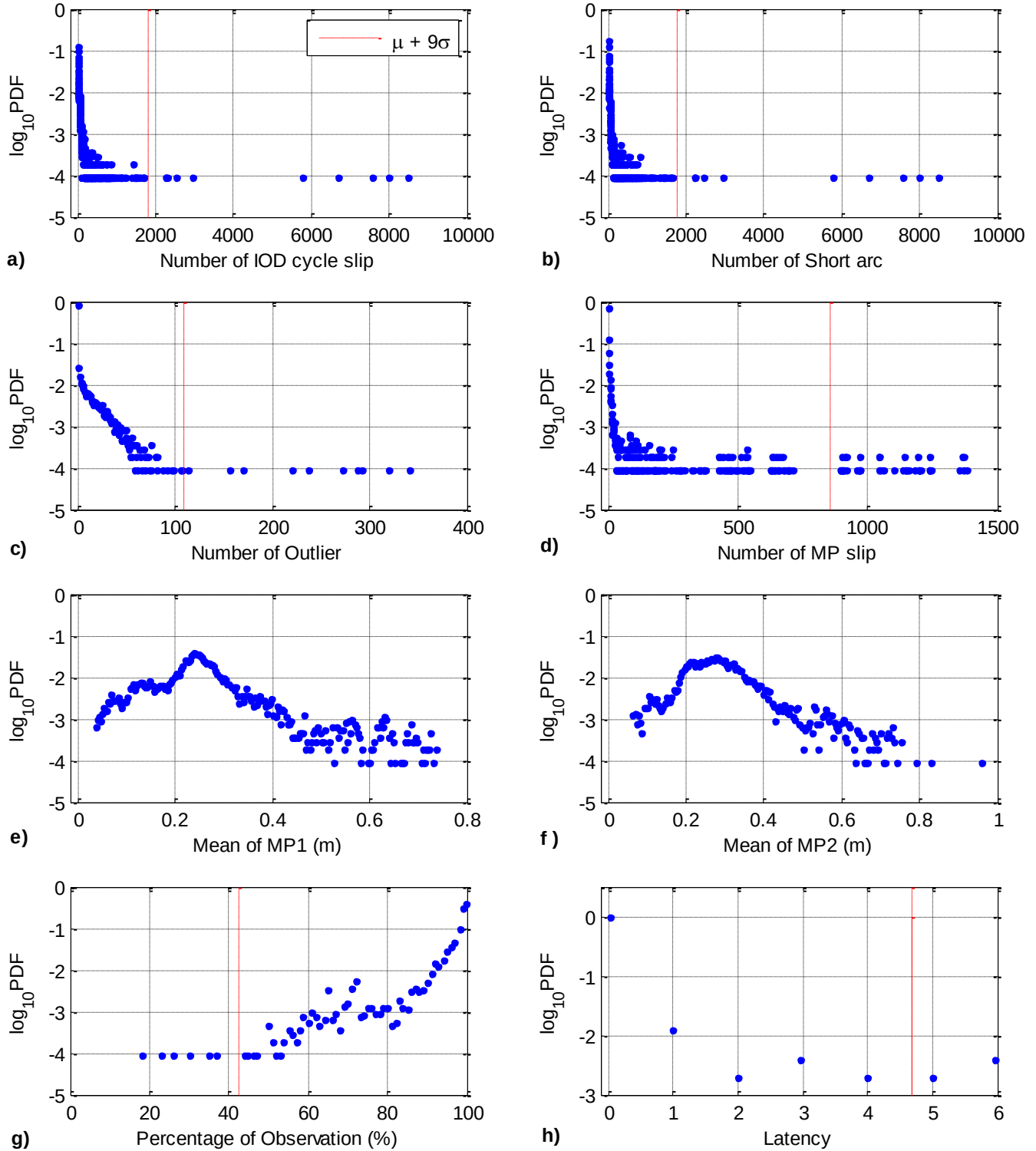


Figure 6. Probability density function of the quality parameters on each station per day, data collected for 7 days: a) number of IOD cycle slips, b) number of short arcs, c) number of outliers, d) number of MP slips, e) mean of MP1, f) mean of MP2, g) percentage of observation, and h) latency.